

Austin Ulrigg

CURRICULUM VITAE

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EDUCATION

University of Washington

B.Sc. in Mathematics, Departmental Honors

- GPA: 3.83/4.00

September 2023 – June 2026

Seattle, WA

PUBLICATIONS / PREPRINTS

The Genus of Bipartite Kneser Graphs

October 2026

Joint work with Alexander Metzger. *Preprint*, *arXiv* submission pending.

Seattle, WA

- Determined the orientable genus of the bipartite Kneser graphs $H(h, 2)$ for every prime $h > 3$ with $h \equiv 3 \pmod{8}$ by constructing vertex-transitive quadrangular embeddings, which attain the Euler lower bound.
- The construction confirms an infinite family of cases of Pisanski's conjecture on quadrilateral embeddings of regular bipartite graphs.

On the Genus Polynomial of Cubic Graphs

July 2026

Independent Research. Submitted to *Discrete Mathematics*. *arXiv*:2607.25410

Seattle, WA

- Proved the genus polynomial of a cubic graph is a cycle matroid invariant and incomparable with the adjacency spectrum, constructed an infinite family of co-spectral pairs whose minimum genera differ.
- Computed the genus polynomials of all 7,875,918 connected cubic graphs through 22 vertices; scripts, run logs, and certificates [archived publicly](#).

An Efficient Genus Algorithm Based on Graph Rotations

Spring 2024–Winter 2025

Joint work with Alexander Metzger

Seattle, WA

- Developed a novel algorithm to compute the orientable genus of graphs in $\mathcal{O}(n(4^m/n)^{n/t})$.
- Successfully determined the exact genus of several graphs for which it was previously unknown.
- Published in *Discrete Mathematics* 349 (2026), 115308. [doi:10.1016/j.disc.2026.115308](https://doi.org/10.1016/j.disc.2026.115308). *arXiv*:2411.07347.
- Integrated into SageMath (`Graph.genus(algorithm='page')`) and used by the House of Graphs database to compute graph genus.

RESEARCH EXPERIENCE

Graph Genera and Minors

Fall 2025

Undergraduate Honors Thesis, Supervisor: François Clément

University of Washington, Seattle

- Authored a comprehensive survey on graph embeddings on surfaces, rotation systems, and genus algorithms (including PAGE).
- Investigated the classification of forbidden toroidal minors and related open problems.
- Full text available: austinulrigg.com/post-thesis.html

Undergraduate Research in Topological Graph Theory

Summer 2025–Winter 2026

Advisor: François Clément

Seattle, WA

- Worked on a project on classifying forbidden minors for the torus, specifically for graphs excluding a K_5 minor.

Washington Experimental Math Lab Researcher

Mar 2024–June 2024; Sep 2024–Dec 2024

Advisors: Professor Hadrian Quan; Professor Stefan Steinerberger

Seattle, WA

- Fall 2024, with Steinerberger: investigated the asymptotic behavior of the dynamical system described in *arXiv*:2409.08961.
- Spring 2024, with Quan: collaborated with a team to investigate wave propagation on graphs, focusing on spectral analysis of the graph Laplacian, graph products, and path homotopy.

PRESENTATIONS

A Practical Genus Algorithm

May 2025; Nov 2025

UW Undergraduate Research Symposium; Northwest Undergraduate Mathematics Symposium (NUMS)

Seattle; Bothell, WA

- Contributed talks (joint work with Alexander Metzger) on a new approach to genus computation using an $\mathcal{O}(n(4^m/n)^{n/t})$ algorithm, emphasizing theoretical advancements and impacts on computational fields.

Graph Genus Algorithm <i>Husky Math Club, University of Washington</i>	Feb 2025 Seattle, WA
Euler Characteristic of a Torus and the Utilities Problem <i>Centralia High School</i>	July 2024 Centralia, WA
Talk for ~30 advanced high-school students on surfaces of various genera, Euler's formula, and gave a solution of the utilities problem on a torus.	

HONORS AND AWARDS

Husky 100 (Nominated by Dept. of Mathematics)	2025
Math Alliance Predoctoral Scholar Nominee	2024
Phi Theta Kappa Honor Society (International Honor Society)	2021–Present
Outstanding Student of the Year Nominee (Centralia College)	2022
High School Valedictorian (Rank 1/200+, Centralia High School)	2022
Merit Scholarships Awarded (Total: \$12,000)	2022

PROFESSIONAL EXPERIENCE

Subject Matter Expert (Contract) <i>Mercor</i>	Aug 2026 – Present Remote
Subject matter expertise for top AI labs through Mercor's expert network; work under NDA.	
Mathematics Domain Expert (AI Training) <i>Outlier AI / DataAnnotation</i>	Aug 2023 – July 2024; Aug 2025 – Present Remote
Author and edit complex mathematical prompts and solutions to refine LLM reasoning capabilities. Develop and apply rubrics to assess model performance and categorize content by difficulty.	

OUTREACH AND SERVICE

Reading Group Organizer <i>University of Washington</i>	2023 – 2025
Co-organized and led reading groups in topology, algebra, linear algebra, and complex analysis for 70+ total students, facilitating discussions, grading, and assigning problems.	
Teaching Assistant (MATH 209) <i>University of Washington</i>	Sep 2024 – Mar 2025
Assist with ODEs/PDEs (heat, wave, etc.); grade assignments, provide one-on-one support.	
Mathematics Tutor (CLUE) <i>University of Washington</i>	Sep 2024 – June 2025
Tutor courses from introductory calculus to abstract algebra.	
Teaching Assistant <i>Husky Math Academy</i>	Oct 2025 – June 2026
Provide instruction, building problem-solving skills beyond the standard curriculum.	
Math Student Council <i>University of Washington</i>	2024 – 2026
Council Member: Organize study halls and events; act as liaison between students and faculty.	
VP & Recruitment Chair <i>Theta Delta Chi Fraternity</i>	2024 – 2026
Elected officer for an 80+ member chapter; oversee recruitment and chapter operations.	
Online Mathematics Community <i>Volunteer Tutor</i>	Jan 2023 – Present
Authored 100,000+ posts in a community of 200k+ members, facilitating math discussions.	
Mathematics Tutor <i>Mathnasium</i>	July 2025 – Sep 2025
Tutored K–12 students, focusing on foundational skills and confidence building.	
Solution Notes: Pearls in Graph Theory <i>Independent Project</i>	May 2024 – June 2024
Authored a 12-page solution set for Hartsfield–Ringel's <i>Pearls in Graph Theory</i>. [View Notes]	
STEM Tutor <i>Centralia College</i>	May 2020 – June 2022
Tutored Calculus I–III, Statistics, and Human Biology; supported center operations.	

GRADUATE COURSEWORK, LANGUAGES, AND INTERESTS

Coursework: Foundations of Combinatorics (Math 563), Modern Algebra I–III (Math 504–506), Linear Algebra (Math 582–583).

Languages: English (native), Spanish (B1), German (A2).

Hobbies: Volleyball (especially beach), Hiking, Dogs, Powerlifting, Reading